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SET - A - 15
JOSÉ RIZAL UNIVERSITY

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EXAMINATION BOOK

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SUBJECT ITC - CSOC 301 i

DO NOT TEAR ANY PAGE OUT OF THIS
EXAMINATION BOOK

TEST 1:

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1. ~~Through~~ With the current situation that we are facing globally, not just the Philippines alone, accurately predicting a Region it is at risk of running out of fuel can provide Actionable Intelligence for Department of Energy and can enable or serve as call to action based on fuel. As by simply looking at Current Inventory, may not yield on an accurate or reliable result. Considering that there are other independent variables or factors that needs to be considered ~~like~~, namely, the Tanker ETA (Hour) or simply the proximity of the oil depot from the region for the tanker to deliver the fuel quickly. In addition, the Daily Average Consumption (Liters), or the demand itself for the motorist of that specific region which is also related to the Vehicle Queue Density. Hence the best course of action (~~info~~) is to ensure that you consider ~~all~~ those independent variables in creating a model and to train, in order to, accurately predict if such region is at risk or if it is high or low.

Current Inventory (Liters) may be pleasing by looking at its ~~(face value)~~ face value although considering that there are region which has low in inventory yet at ~~(a low)~~ tagged as Low Risk, in terms of Fuel-Risk. (PH-008)

a. Since the nature or the data type of the Target Attribute is Categorical, the logical option to choose is Logistic Regression. As Linear Regression is designed ideally to be used to output continuous ~~numerical~~ and ~~any~~ numerical value, in contrast the Logistic Regression, is the preferred option as it is designed to output a value from 0 to 1, while having the decision threshold as the decision maker (default to 0.5) to decide if the value is higher, hence it will be predicted as High ~~risk~~ on the other hand if the value is lower than 0.5 it will be predicted as low risk.

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based on what?

interpret?

3. the raw output of 0.82 will be interpreted as High, as by following the default value for the decision threshold (0.5), any value higher than the decision threshold value will be interpreted as High, unless the decision threshold is altered. Hence since PH-001 is considered or tagged as High Risk, ~~may consider~~ the Department of Energy can allocate more fuel to meet the demand based on the Daily Average consumption and vehicle Avere Density. In addition, may consider to allocate more Tankers to deliver the fuel or find and optimize the route for the fuel to be delivered lesser than the ETA.

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TEST II:

1. Having a small sample size for a dataset is considered detrimental in ~~(training)~~ training and testing the model, especially if you consider utilizing 70/30 train-test split as by using scikit ~~(learn library)~~ learn library it shuffles the dataset and select 70% of the whole dataset and utilize it for ~~test~~ training and the other 30% for testing. Hence, this may involve the luck of the draw in selecting that 70% ~~and~~ and ~~may~~ considered risky if the dataset is imbalanced which is true in real-world scenarios. In addition if for some reason the ~~training~~ rows allocated by the model is highly imbalanced, the model will have difficulties in predicting whether it is high risk or low. ~~(also)~~ through utilizing the 40 rows the 27 rows are allocated for testing which leaves only 12 rows to be used for testing. Hence 5-fold cross validation is a better option.

7. Based on my understanding, the 40 rows will be divided to 5 which leaves 2 per fold. and will proceed testing ~~and training~~ the model 5 times and the 2 will be used for training while 32 is utilize for testing per iteration.

sample diagram:

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	32	33	34	35
36	37	38	39	40		

wrong diagram!

the numbers encircled will be utilized for testing while the remaining is for training. and the order will be changed per iteration ~~except that~~ until it reaches 5 iteration and the mean of all iterations will be measured.

2. Relying on the model as the ~~decision~~ ~~marker~~ as a guide in making a ~~decision~~ decision in dispatching limited tankers. May result in oversupply on a specific region ~~while~~ and the worst is, other regions will be scarce or undersupply if for some reason the model predicted False Negative which is not a best scenario to imagine if that happens to the motorists for that region. Hence Ensuring to lead towards checking the Recall will ensure that the model is trained well or may require improvement by changing the parameters of proper data cleaning and preparation.

$$\text{Recall} = \frac{TP}{TP + FN}$$

why oversupply when it is predicted as low risk?